牙医、资质与治疗目录

DENTISTS（牙医）

PK	id	TEXT	NN
	employee_code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	title_en	TEXT	NN
	title_zh	TEXT	NN
	email	TEXT	NN IDX
	phone	TEXT	NN
	hire_date	DATE	NN
	employment_status	TEXT	NN IDX
	languages_json	TEXT	NN
	biography_en	TEXT	NN
	biography_zh	TEXT	NN
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

QUALIFICATION_TYPES（资质类型）

PK	id	TEXT	NN
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	category	TEXT	NN IDX
	description_en	TEXT	NN
	description_zh	TEXT	NN
	status	TEXT	NN IDX
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

DENTIST_QUALIFICATIONS（牙医资质证书）

PK	id	TEXT	NN
FK	dentist_id → dentists	TEXT	NN IDX
FK	qualification_type_id → qualification_typ…	TEXT	NN IDX
	issuing_body	TEXT	NN
	reference_number	TEXT	NN
	issued_on	TEXT	NN
	expires_on	TEXT	NULL IDX
	verification_status	TEXT	NN IDX
	verified_at	INTEGER	NN
	status	TEXT	NN IDX
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

DENTIST_CLINIC_ASSIGNMENTS（牙医诊所任职）

PK	id	TEXT	NN
FK	dentist_id → dentists	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NN IDX
	is_primary	INTEGER	NN DEF
	effective_from	TEXT	NN
	effective_to	TEXT	NULL
	weekly_hours	INTEGER	NN DEF
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF

TREATMENT_CATALOG（治疗项目目录）

PK	id	TEXT	NN
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	description_en	TEXT	NN
	description_zh	TEXT	NN
	duration_minutes	INTEGER	NN
	standard_price_fen	INTEGER	NN
	estimated_cost_fen	INTEGER	NN
	version	INTEGER	NN IDX DEF
	effective_from	TEXT	NN
	effective_to	TEXT	NULL
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

TREATMENT_QUALIFICATION_REQUIREMENTS（治疗资质要求）

PK/FK	treatment_id → treatment_catalog	TEXT	NN
PK/FK	qualification_type_id → qualification_typ…	TEXT	NN IDX
	requirement_level	TEXT	NN DEF
	created_at	TEXT	NN DEF

排班、资源与预约

CLINIC_CHAIRS（诊所牙椅）

PK	id	TEXT	NN
FK	clinic_id → clinics	TEXT	NN IDX
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	room_name_en	TEXT	NN
	room_name_zh	TEXT	NN
	equipment_json	TEXT	NN
	is_accessible	INTEGER	NN DEF
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

CLINIC_OPERATING_HOURS（诊所营业时间）

PK/FK	clinic_id → clinics	TEXT	NN
PK	day_of_week	INTEGER	NN
	opens_at	TEXT	NN
	closes_at	TEXT	NN
	slot_minutes	INTEGER	NN DEF
	is_closed	INTEGER	NN DEF
	updated_at	TEXT	NN DEF

CLINIC_CLOSURES（诊所停诊安排）

PK	id	TEXT	NN
FK	clinic_id → clinics	TEXT	NN IDX
	starts_at	TEXT	NN IDX
	ends_at	TEXT	NN IDX
	reason_code	TEXT	NN
	reason_en	TEXT	NN
	reason_zh	TEXT	NN
	created_at	TEXT	NN DEF

APPOINTMENTS（预约）

PK	id	TEXT	NN
	appointment_code	TEXT	NN IDX
FK	patient_id → patients	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NN IDX
FK	dentist_id → dentists	TEXT	NN IDX
FK	chair_id → clinic_chairs	TEXT	NULL IDX
FK	treatment_id → treatment_catalog	TEXT	NN
	scheduled_start_at	TEXT	NN IDX
	scheduled_end_at	TEXT	NN IDX
	duration_minutes	INTEGER	NN
	status	TEXT	NN IDX DEF
	booking_channel	TEXT	NN DEF
	priority	TEXT	NN DEF
	confirmation_state	TEXT	NN DEF
	risk_flag	TEXT	NN DEF
	notes	TEXT	NULL
	cancellation_reason_code	TEXT	NULL
	appointment_version	INTEGER	NN DEF
FK	created_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

DENTIST_AVAILABILITY_RULES（牙医常规可用规则）

PK	id	TEXT	NN
FK	dentist_id → dentists	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NN IDX
	day_of_week	INTEGER	NN IDX
	starts_at	TEXT	NN
	ends_at	TEXT	NN
	effective_from	TEXT	NN
	effective_to	TEXT	NULL
	recurrence_rule	TEXT	NN DEF
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF

DENTIST_SCHEDULE_EXCEPTIONS（牙医排班例外）

PK	id	TEXT	NN
FK	dentist_id → dentists	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NULL IDX
	starts_at	TEXT	NN
	ends_at	TEXT	NN
	exception_type	TEXT	NN
	reason_en	TEXT	NULL
	reason_zh	TEXT	NULL
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF

APPOINTMENT_STATUS_HISTORY（预约状态历史）

PK	id	TEXT	NN
FK	appointment_id → appointments	TEXT	NN IDX
	from_status	TEXT	NULL
	to_status	TEXT	NN
	reason_code	TEXT	NULL
	reason_text	TEXT	NULL
FK	actor_user_id → app_users	TEXT	NULL IDX
	occurred_at	TEXT	NN IDX
	metadata_json	TEXT	NN DEF
	created_at	TEXT	NN DEF

WAITLIST_ENTRIES（候补登记）

PK	id	TEXT	NN
FK	patient_id → patients	TEXT	NN IDX
FK	treatment_id → treatment_catalog	TEXT	NN IDX
FK	preferred_clinic_id → clinics	TEXT	NULL IDX
	earliest_date	TEXT	NULL
	latest_date	TEXT	NULL
	preferred_periods_json	TEXT	NN DEF
	acceptable_clinic_ids_json	TEXT	NN DEF
	notice_hours	INTEGER	NN DEF
	priority_score	INTEGER	NN DEF
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

REQUESTED_TREATMENT_ITEMS（预约治疗诉求）

PK	id	TEXT	NN
FK	plan_id → treatment_plans	TEXT	NN IDX
FK	visit_id → clinical_visits	TEXT	NN IDX
FK	treatment_id → treatment_catalog	TEXT	NN
	sequence	INTEGER	NN IDX
	tooth_code	TEXT	NULL
	surfaces_json	TEXT	NN DEF
	units	INTEGER	NN DEF
	priority	TEXT	NN DEF
	rationale_en	TEXT	NN
	rationale_zh	TEXT	NN DEF
	status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF

WAITLIST_OFFERS（候补邀约）

PK	id	TEXT	NN
FK	waitlist_entry_id → waitlist_entries	TEXT	NN IDX
FK	released_appointment_id → appointments	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NN
FK	dentist_id → dentists	TEXT	NN
FK	chair_id → clinic_chairs	TEXT	NULL
	offered_start_at	TEXT	NN
	offered_end_at	TEXT	NN
	compatibility_score	INTEGER	NN
	reasons_json	TEXT	NN
	status	TEXT	NN IDX DEF
	expires_at	TEXT	NN IDX
	responded_at	TEXT	NULL
	created_at	TEXT	NN DEF

临床就诊与病历版本

CLINICAL_VISITS（临床就诊）

PK	id	TEXT	NN
	visit_code	TEXT	NN IDX
FK	appointment_id → appointments	TEXT	NN IDX
FK	patient_id → patients	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NN IDX
FK	dentist_id → dentists	TEXT	NN IDX
FK	chair_id → clinic_chairs	TEXT	NULL
	started_at	TEXT	NN IDX
	chief_complaint_en	TEXT	NN DEF
	chief_complaint_zh	TEXT	NN DEF
	safety_gate_status	TEXT	NN DEF
	ended_at	TEXT	NULL
	completed_at	TEXT	NULL
	visit_version	INTEGER	NN DEF
FK	created_by_user_id → app_users	TEXT	NULL
	updated_at	TEXT	NN DEF

VISIT_ALLERGY_ACKNOWLEDGEMENTS（就诊过敏复核）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
FK	allergy_id → patient_allergies	TEXT	NN IDX
FK	acknowledged_by_user_id → app_users	TEXT	NN
	acknowledged_at	TEXT	NN
	precaution_en	TEXT	NN
	precaution_zh	TEXT	NN
	outcome	TEXT	NN DEF
	severity	TEXT	NN DEF
	finding_en	TEXT	NN
	finding_zh	TEXT	NN
	status	TEXT	NN DEF
	observed_by_dentist_id → dentists	TEXT	NN
	observed_at	TEXT	NN
	supersedes_observation_id	TEXT	NULL
	created_at	TEXT	NN DEF

ODONTOGRAM_OBSERVATIONS（牙位观察）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
FK	patient_id → patients	TEXT	NN IDX
	tooth_code	TEXT	NN
	dentition	TEXT	NN IDX
	surfaces_json	TEXT	NN
	observation_type	TEXT	NN
	severity	TEXT	NN
	finding_en	TEXT	NN
	finding_zh	TEXT	NN
	status	TEXT	NN DEF
	observed_by_dentist_id → dentists	TEXT	NN
	observed_at	TEXT	NN
	supersedes_observation_id	TEXT	NULL
	created_at	TEXT	NN DEF

CLINICAL_NOTES（临床记录）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
	note_type	TEXT	NN DEF
	current_version	INTEGER	NN DEF
	status	TEXT	NN DEF
FK	signed_by_dentist_id → dentists	TEXT	NULL
	signed_at	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

TREATMENT_PLANS（治疗计划）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
	plan_code	TEXT	NN IDX
	status	TEXT	NN IDX
	clinical_priority	TEXT	NN DEF
	estimated_total_fen	INTEGER	NN DEF
	acceptance_status	TEXT	NN DEF
	accepted_at	TEXT	NULL
	accepted_by	TEXT	NULL
	valid_until	TEXT	NULL
	notes_en	TEXT	NN
	notes_zh	TEXT	NN
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

CLINICAL_ATTACHMENTS（临床附件）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
	category	TEXT	NN
	file_name	TEXT	NN
	mime_type	TEXT	NN
	storage_key	TEXT	NN IDX
	size_bytes	INTEGER	NN
	description_en	TEXT	NN
	description_zh	TEXT	NN
	content_checksum	TEXT	NN
	captured_at	TEXT	NN
	status	TEXT	NN DEF
	created_at	TEXT	NN DEF

CLINICAL_NOTE_VERSIONS（临床记录版本）

PK	id	TEXT	NN
FK	clinical_note_id → clinical_notes	TEXT	NN IDX
	version_number	INTEGER	NN
	content_en	TEXT	NN
	content_zh	TEXT	NN
	amendment_reason	TEXT	NN
FK	author_user_id → app_users	TEXT	NN
	content_checksum	TEXT	NN
	created_at	TEXT	NN DEF

ACTUAL_TREATMENT_ITEMS（实际治疗项目）

PK	id	TEXT	NN
FK	visit_id → clinical_visits	TEXT	NN IDX
FK	requested_treatment_item_id → requested_t…	TEXT	NULL IDX
FK	treatment_id → treatment_catalog	TEXT	NN
FK	provider_dentist_id → dentists	TEXT	NN
	tooth_code	TEXT	NULL
	surfaces_json	TEXT	NN
	units	INTEGER	NN DEF
	effective_unit_price_fen	INTEGER	NN
	line_amount_fen	INTEGER	NN
	price_source	TEXT	NN
	price_override_reason	TEXT	NULL
	notes_en	TEXT	NN
	notes_zh	TEXT	NN
	performed_at	TEXT	NN
	status	TEXT	NN DEF
	created_at	TEXT	NN DEF

结算、收款与成本

FINANCIAL_REPORTING_PERIODS（财务报告期间）

PK	id	TEXT	NN
	period_code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	starts_on	TEXT	NN IDX
	ends_on	TEXT	NN IDX
	status	TEXT	NN IDX DEF
	close_checklist_json	TEXT	NN DEF
	locked_at	TEXT	NULL
FK	locked_by_user_id → app_users	TEXT	NULL
	reopened_at	TEXT	NULL
FK	reopened_by_user_id → app_users	TEXT	NULL
	reopen_reason	TEXT	NULL
	period_version	INTEGER	NN DEF
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

INVOICES（诊疗结算单）

PK	id	TEXT	NN
	invoice_code	TEXT	NN IDX
FK	visit_id → clinical_visits	TEXT	NN IDX
FK	patient_id → patients	TEXT	NN
FK	clinic_id → clinics	TEXT	NN IDX
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
	currency	TEXT	NN DEF
	subtotal_fen	INTEGER	NN DEF
	adjustment_total_fen	INTEGER	NN DEF
	tax_total_fen	INTEGER	NN DEF
	total_fen	INTEGER	NN DEF
	paid_fen	INTEGER	NN DEF
	refunded_fen	INTEGER	NN DEF
	balance_fen	INTEGER	NN IDX DEF
	issued_at	TEXT	NULL
	due_at	TEXT	NULL IDX
	voided_at	TEXT	NULL
	void_reason	TEXT	NULL
	invoice_version	INTEGER	NN DEF
FK	created_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

INVOICE_LINES（结算明细）

PK	id	TEXT	NN
FK	invoice_id → invoices	TEXT	NN IDX
FK	actual_treatment_item_id → actual_treatme…	TEXT	NN
	line_number	INTEGER	NN IDX
	treatment_code_snapshot	TEXT	NN
	description_en_snapshot	TEXT	NN
	description_zh_snapshot	TEXT	NN
	tooth_code_snapshot	TEXT	NULL
	surfaces_json_snapshot	TEXT	NN DEF
	units_snapshot	TEXT	NN DEF
	unit_price_fen_snapshot	INTEGER	NN
	amount_fen_snapshot	INTEGER	NN
	cost_fen_snapshot	INTEGER	NN DEF
	performed_at_snapshot	TEXT	NN
	created_at	TEXT	NN DEF

INVOICE_ADJUSTMENTS（结算调整）

PK	id	TEXT	NN
FK	invoice_id → invoices	TEXT	NN IDX
	adjustment_type	TEXT	NN
	amount_fen	INTEGER	NN
	reason_code	TEXT	NN
	reason_text	TEXT	NN
	status	TEXT	NN IDX DEF
FK	authored_by_user_id → app_users	TEXT	NULL
FK	approved_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

PAYMENTS（收款流水）

PK	id	TEXT	NN
	payment_code	TEXT	NN IDX
FK	invoice_id → invoices	TEXT	NN IDX
	amount_fen	INTEGER	NN
	method	TEXT	NN
	status	TEXT	NN IDX
	simulated_gateway_reference	TEXT	NULL
	received_at	TEXT	NN IDX
FK	recorded_by_user_id → app_users	TEXT	NULL
	notes	TEXT	NN DEF
	created_at	TEXT	NN DEF

PAYMENT_REFUNDS（退款流水）

PK	id	TEXT	NN
	refund_code	TEXT	NN IDX
FK	payment_id → payments	TEXT	NN IDX
	amount_fen	INTEGER	NN
	reason_code	TEXT	NN
	reason_text	TEXT	NN
	status	TEXT	NN IDX DEF
FK	authorized_by_user_id → app_users	TEXT	NULL
	refunded_at	TEXT	NN
	created_at	TEXT	NN DEF

TREATMENT_COST_ENTRIES（治疗成本记录）

PK	id	TEXT	NN
FK	actual_treatment_item_id → actual_treatme…	TEXT	NN IDX
FK	visit_id → clinical_visits	TEXT	NN
FK	clinic_id → clinics	TEXT	NN IDX
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
	catalog_estimated_cost_fen	INTEGER	NN
	materials_cost_fen	INTEGER	NN DEF
	laboratory_cost_fen	INTEGER	NN DEF
	provider_cost_fen	INTEGER	NN DEF
	total_cost_fen	INTEGER	NN
	cost_date	TEXT	NN
	source	TEXT	NN DEF
	created_at	TEXT	NN DEF

期间核算与激励

OPERATING_EXPENSES（运营费用）

PK	id	TEXT	NN
FK	clinic_id → clinics	TEXT	NN IDX
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
	category	TEXT	NN IDX
	vendor_name	TEXT	NN DEF
	amount_fen	INTEGER	NN
	incurred_on	TEXT	NN IDX
	description_en	TEXT	NN
	description_zh	TEXT	NN
	status	TEXT	NN DEF
FK	entered_by_user_id → app_users	TEXT	NULL
FK	approved_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

FINANCIAL_CLOSE_EXCEPTIONS（期间关闭异常）

PK	id	TEXT	NN
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NULL
	exception_type	TEXT	NN
	severity	TEXT	NN IDX
	entity_type	TEXT	NN
	entity_id	TEXT	NN IDX
	description_en	TEXT	NN
	description_zh	TEXT	NN
	status	TEXT	NN IDX DEF
	resolved_at	TEXT	NULL
	resolution_note	TEXT	NULL
	created_at	TEXT	NN DEF

FINANCIAL_PERIOD_RESULTS（期间结算结果）

PK	id	TEXT	NN
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NULL IDX
	scope_key	TEXT	NN IDX
	recognized_revenue_fen	INTEGER	NN
	treatment_cost_fen	INTEGER	NN
	operating_expense_fen	INTEGER	NN
	net_profit_fen	INTEGER	NN IDX
	net_margin_basis_points	INTEGER	NN
	invoice_count	INTEGER	NN
	open_exception_count	INTEGER	NN DEF
	calculation_inputs_json	TEXT	NN
	status	TEXT	NN IDX DEF
	calculated_at	TEXT	NULL
	approved_at	TEXT	NULL
FK	approved_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

INCENTIVE_POOLS（激励池）

PK	id	TEXT	NN
FK	reporting_period_id → financial_reporting…	TEXT	NN IDX
FK	network_result_id → financial_period_resu…	TEXT	NN
	rate_basis_points	INTEGER	NN DEF
	eligible_net_profit_fen	INTEGER	NN
	pool_amount_fen	INTEGER	NN
	formula_evidence	TEXT	NN
	status	TEXT	NN DEF
	approved_at	TEXT	NULL
FK	approved_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

扩张情景与候选诊所

EXPANSION_SCENARIOS（扩张情景）

PK	id	TEXT	NN
	scenario_code	TEXT	NN IDX
	version	INTEGER	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	description_en	TEXT	NN
	description_zh	TEXT	NN
	forecast_horizon_months	INTEGER	NN DEF
	status	TEXT	NN IDX DEF
	advisory_disclaimer_en	TEXT	NN
	advisory_disclaimer_zh	TEXT	NN
FK	created_by_user_id → app_users	TEXT	NULL
FK	approved_by_user_id → app_users	TEXT	NULL
	approved_at	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN IDX DEF

EXPANSION_CANDIDATE_CLINICS（候选诊所）

PK	id	TEXT	NN
FK	scenario_id → expansion_scenarios	TEXT	NN IDX
	candidate_code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	city_en	TEXT	NN
	city_zh	TEXT	NN
	district_en	TEXT	NN
	district_zh	TEXT	NN
	planned_chair_count	INTEGER	NN
	planned_dentist_count	INTEGER	NN
	planned_support_staff_count	INTEGER	NN
	forecast_quarterly_appointments	INTEGER	NN
	initial_capital_fen	INTEGER	NN
	quarterly_fixed_cost_fen	INTEGER	NN
	required_qualifications_json	TEXT	NN DEF
	qualification_coverage_basis_points	INTEGER	NN
	travel_constraint_minutes	INTEGER	NN
	readiness_status	TEXT	NN IDX DEF
	created_at	TEXT	NN DEF

EXPANSION_ASSUMPTIONS（扩张假设）

PK	id	TEXT	NN
FK	scenario_id → expansion_scenarios	TEXT	NN IDX
	assumption_key	TEXT	NN IDX
	label_en	TEXT	NN
	label_zh	TEXT	NN
	value_text	TEXT	NN
	unit	TEXT	NN
	source_en	TEXT	NN
	source_zh	TEXT	NN
	created_at	TEXT	NN DEF

EXPANSION_SCENARIO_RESULTS（扩张情景结果）

PK	id	TEXT	NN
FK	scenario_id → expansion_scenarios	TEXT	NN IDX
	projected_quarter_revenue_fen	INTEGER	NN
	projected_treatment_cost_fen	INTEGER	NN
	projected_operating_expense_fen	INTEGER	NN
	projected_net_profit_fen	INTEGER	NN
	projected_margin_basis_points	INTEGER	NN
	payback_months	INTEGER	NN
	demand_coverage_basis_points	INTEGER	NN
	qualification_coverage_basis_points	INTEGER	NN
	composite_score	INTEGER	NN IDX
	risk_level	TEXT	NN IDX
	formula_evidence_json	TEXT	NN
	calculated_at	TEXT	NN
	created_at	TEXT	NN DEF

患者、预约与临床

ORGANIZATIONS（组织）

PK	id	TEXT	NN
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	timezone	TEXT	NN DEF
	currency	TEXT	NN DEF
	status	TEXT	NN IDX
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

CLINICS（诊所）

PK	id	TEXT	NN
FK	organization_id → organizations	TEXT	NN IDX
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	city_en	TEXT	NN
	city_zh	TEXT	NN
	address_en	TEXT	NN
	address_zh	INTEGER	NULL
	phone	TEXT	NN
	email	TEXT	NULL
	chair_capacity	INTEGER	NN DEF
	status	TEXT	NN IDX DEF
	opened_on	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

APP_USERS（系统账户）

PK	id	TEXT	NN
	auth_subject	TEXT	NN IDX
	email	TEXT	NN IDX
	display_name	TEXT	NN
	phone_e164	TEXT	NN
	username	TEXT	NULL IDX
	password_hash	TEXT	NULL
	password_salt	TEXT	NULL
	password_iterations	INTEGER	NN
	password_set_at	TEXT	NULL
	last_auth_method	TEXT	NULL
	failed_login_count	INTEGER	NN DEF
	locked_until	TEXT	NULL
	preferred_language	TEXT	NN
	status	TEXT	NN IDX DEF
	last_login_at	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

ACCOUNT_INVITATIONS（账户邀请）

PK	id	TEXT	NN
	phone_e164	TEXT	NN IDX
FK	target_user_id → app_users	TEXT	NULL
	role_id	TEXT	NULL
	clinic_id	TEXT	NULL
	status	TEXT	NN IDX DEF
	expires_at	TEXT	NN
	accepted_at	TEXT	NULL
FK	created_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

ROLES（角色）

PK	id	TEXT	NN
	code	TEXT	NN IDX
	name_en	TEXT	NN
	name_zh	TEXT	NN
	description_en	TEXT	NN
	description_zh	TEXT	NN
	is_system	INTEGER	NN DEF
	status	TEXT	NN DEF
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

ROLE_PERMISSIONS（角色权限）

PK/FK	role_id → roles	TEXT	NN
PK/FK	permission_id → permissions	TEXT	NN IDX
	created_at	TEXT	NN DEF

AUDIT_EVENTS（审计事件）

PK	id	INTEGER	NN
	request_id	TEXT	NN IDX
FK	actor_user_id → app_users	TEXT	NULL IDX
	actor_subject	TEXT	NULL
	action	TEXT	NN IDX
	entity_type	TEXT	NN IDX
	entity_id	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NULL IDX
	outcome	TEXT	NN DEF
	source	TEXT	NN DEF
	reason	TEXT	NULL
	before_json	TEXT	NULL
	after_json	TEXT	NULL
	metadata_json	TEXT	NN DEF
	occurred_at	TEXT	NN IDX DEF

SYSTEM_SETTINGS（系统配置）

PK/FK	organization_id → organizations	TEXT	NN
PK	setting_key	TEXT	NN IDX
	value_json	TEXT	NN
FK	updated_by_user_id → app_users	TEXT	NULL
	updated_at	TEXT	NN DEF

AUTH_IDENTITIES（外部身份）

PK	id	TEXT	NN
FK	user_id → app_users	TEXT	NN IDX
	provider	TEXT	NN IDX
	provider_subject	TEXT	NN
	profile_json	TEXT	NN
	last_used_at	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

AUTH_OAUTH_STATES（OAuth 状态）

PK	id	TEXT	NN
	provider	TEXT	NN
	state_hash	TEXT	NN IDX
	return_to	TEXT	NN
	expires_at	TEXT	NN IDX
	consumed_at	TEXT	NULL
	created_at	TEXT	NN DEF

AUTH_OTP_CHALLENGES（验证码挑战）

PK	id	TEXT	NN
	phone_e164	TEXT	NN IDX
	purpose	TEXT	NN IDX
	code_hash	TEXT	NN
	code_salt	TEXT	NN
	expires_at	TEXT	NN IDX
	consumed_at	TEXT	NULL
	attempt_count	INTEGER	NN DEF
	max_attempts	INTEGER	NN DEF
	request_ip_hash	TEXT	NULL
	created_at	TEXT	NN IDX DEF

AUTH_SESSIONS（认证会话）

PK	id	TEXT	NN
FK	user_id → app_users	TEXT	NN IDX
	token_hash	TEXT	NN IDX
	auth_method	TEXT	NN
	expires_at	TEXT	NN IDX
	revoked_at	TEXT	NULL
	last_seen_at	TEXT	NULL
	ip_hash	TEXT	NULL
	user_agent_hash	TEXT	NULL
	created_at	TEXT	NN DEF

PATIENT_ACCOUNT_LINKS（患者账户绑定）

PK	id	TEXT	NN
FK	user_id → app_users	TEXT	NN IDX
FK	patient_id → patients	TEXT	NN IDX
	verification_method	TEXT	NN
	verified_at	TEXT	NN
	created_at	TEXT	NN DEF

NOTIFICATION_EVENTS（通知事件）

PK	id	TEXT	NN
FK	appointment_id → appointments	TEXT	NULL IDX
	channel	TEXT	NN
	template_code	TEXT	NN
	recipient_masked	TEXT	NN
	locale	TEXT	NN DEF
	status	TEXT	NN IDX DEF
	idempotency_key	TEXT	NN IDX
	queued_at	TEXT	NN
	sent_at	TEXT	NULL
	delivered_at	TEXT	NULL
	failure_code	TEXT	NULL
	created_at	TEXT	NN DEF

PERMISSIONS（权限）

PK	id	TEXT	NN
	code	TEXT	NN IDX
	domain	TEXT	NN IDX
	description_en	TEXT	NN
	description_zh	TEXT	NN
	created_at	TEXT	NN DEF

USER_ROLE_ASSIGNMENTS（用户角色授权）

PK	id	TEXT	NN
FK	user_id → app_users	TEXT	NN
FK	role_id → roles	TEXT	NN
FK	clinic_id → clinics	TEXT	NULL IDX
	status	TEXT	NN IDX DEF
	effective_from	TEXT	NN
	effective_to	TEXT	NULL
FK	granted_by_user_id → app_users	TEXT	NULL
	created_at	TEXT	NN DEF

PATIENTS（患者）

PK	id	TEXT	NN
	patient_code	TEXT	NN IDX
	name_en	TEXT	NN IDX
	name_zh	TEXT	NN
	date_of_birth	TEXT	NN
	gender	TEXT	NN
	phone	TEXT	NN IDX
	email	TEXT	NN IDX
	government_id_last4	TEXT	NULL
	preferred_language	TEXT	NN DEF
	communication_channel	TEXT	NN DEF
FK	primary_clinic_id → clinics	TEXT	NULL IDX
	status	TEXT	NN IDX DEF
	last_visit_at	TEXT	NULL
	created_at	TEXT	NN DEF
	updated_at	TEXT	NN DEF

PATIENT_TIMELINE_EVENTS（患者时间线事件）

PK	id	TEXT	NN
FK	patient_id → patients	TEXT	NN IDX
FK	clinic_id → clinics	TEXT	NULL
	event_type	TEXT	NN IDX
	occurred_at	TEXT	NN IDX
	title_en	TEXT	NN
	title_zh	TEXT	NN
	description_en	TEXT	NN
	description_zh	TEXT	NN
	reference_type	TEXT	NULL IDX
	reference_id	TEXT	NULL IDX
	status	TEXT	NN
	amount_fen	INTEGER	NULL
	metadata_json	TEXT	NN DEF
	created_at	TEXT	NN DEF

EMERGENCY_CONTACTS（紧急联系人）

PK	id	TEXT	NN
FK	patient_id → patients	TEXT	NN IDX
	contact_name	TEXT	NN
	relationship_en	TEXT	NN
	relationship_zh	TEXT	NN
	phone	TEXT